

Variability in morphology and immunohistochemistry of Crohn's disease-associated small bowel neoplasms: implications of Claudin 18 and Cadherin 17 expression for tumor-targeted immunotherapies.

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Inflammatory bowel disease-associated colorectal carcinomas are known to have different morphology, immunoprofile, and genetic findings from sporadic colorectal carcinomas; however, little is known for Crohn's disease-associated small bowel neoplasms (CD-SBNs). Cadherin 17 is a useful biomarker of adenocarcinomas with intestinal phenotype and recently reported as an ideal target for chimeric antigen receptor T-cells (CAR-T) therapy for gastrointestinal carcinoma. Claudin 18 is a cell adhesion protein, and Claudin18 isoform 2 (CLDN18.2) is frequently expressed at high levels in gastric-type adenocarcinoma. Zolbetuximab, a targeted monoclonal antibody, has been developed for CLDN18.2-positive gastroesophageal adenocarcinoma. We examined a series of CD-SBNs for both Cadherin 17 and Claudin 18, and also hypothesized that expression of Claudin 18 was associated with gastric phenotype. We performed histological and immunohistochemical examinations on 25 CD-SBNs. Most of adenocarcinomas showed tubular morphology as seen in gastric carcinomas, whereas a subset of dysplasia was morphologically similar to that of the large bowel. Cadherin17 and Claudin 18 expression was identified in 93% and 57% CD-associated adenocarcinomas respectively. In Cadherin 17-positive CD-SBNs, frequent MUC5AC, MUC6, and Claudin18 expression was identified (61%, 57%, and 57%, respectively). Claudin 18-positive CD-SBNs showed significantly more MUC5AC and MUC6 expression than Claudin 18-negative CD-SBNs ($P = 0.005$, < 0.001 respectively). In CD-associated small bowel adenocarcinomas, Cadherin 17 expression was frequently retained and Claudin 18 was frequently co-expressed. Claudin 18 had a positive correlation with the expression of gastric mucins. These results suggest that CD-associated small bowel adenocarcinomas may be candidates for Cadherin 17- and Claudin 18-targeted immunotherapies.

Preparation of Radiolabeled Zolbetuximab Targeting CLDN18.2 and Its Preliminary Evaluation for Potential Clinical Applications.

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Claudin18.2 (CLDN18.2), due to its high expression in various gastric cancer tissues, is considered an optimal target for antitumor drug molecules. In this study, we obtained the labeled compounds of [125I]I-zolbetuximab using the Iodogen method. Under the optimum labeling conditions, the molar activity of [125I]I-zolbetuximab was 1.75×10^2 GBq/ μ mol, and the labeling efficiency was more than 99%. The labeled compounds exhibited excellent in vitro stability in both phosphate buffer saline (PBS, pH = 7.4) and fetal bovine serum systems (FBS) (radiochemical purity $>90\%$ at 72 h). The uptake percentage of [125I]I-zolbetuximab in MKN45-CLDN18.2 cells is $24.69 \pm 0.84\%$ after 6 h. The saturation binding assay and specificity assay further demonstrated the high specificity of [125I]I-zolbetuximab for CLDN18.2. The long retention at the tumor site and rapid metabolic clearance at other organ sites of [125I]I-zolbetuximab were observed in small-animal SPECT-CT imaging. The same trend was also observed in the biodistribution study. Due to the excellent targeting ability of zolbetuximab for CLDN18.2, [125I]I-zolbetuximab exhibits strong specific binding and retention with cells and tumors highly expressing CLDN18.2. However, the balance between mAb's longer cycle time in vivo and targeting binding and retention ability should be intensively considered for using this kind of radiopharmaceutical in the diagnosis and treatment of CLDN18.2-positive gastric cancer.

The preclinical discovery and development of zolbetuximab for the treatment of gastric cancer.

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Gastric cancer remains a formidable challenge in oncology with high mortality rates and few advancements in treatment. Claudin-18.2 (CLDN18.2) is a tight junction protein primarily expressed in the stomach and is frequently overexpressed in certain subsets of gastric cancers. Targeting CLDN18.2 with monoclonal antibodies, such as zolbetuximab (IMAB362), has shown promising efficacy results in combination with chemotherapy. The molecular cell biology of CLDN18.2 is discussed along with studies demonstrating the utility of CLDN18.2 expression as a biomarker and therapeutic target. Important clinical studies are reviewed, including Phase III trials, SPOTLIGHT and GLOW, which demonstrate the efficacy of zolbetuximab in combination with chemotherapy in patients with CLDN18.2-positive advanced gastric cancer. CLDN18.2 is involved in gastric differentiation through maintenance of epithelial barrier function and coordination of signaling pathways, and its expression in gastric cancers reflects a 'gastric differentiation' program. Targeting Claudin-18.2 represents the first gastric cancer specific 'targeted' treatment. Further studies are needed to determine its role within current gastric cancer treatment sequencing, including HER2-targeted therapies and immunotherapies. Management strategies will also be needed to better mitigate zolbetuximab-related treatment side effects, including gastrointestinal (GI) toxicities.

Progress of Clinical Studies Targeting Claudin18.2 for the Treatment of Gastric Cancer.

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Claudin18.2 is a tight junction protein, highly selective, generally expressed only in normal gastric mucosal epithelial cells, which can effectively maintain the polarity of epithelial and endothelial cells, thus effectively regulating the permeability and conductance of the paracellular pathway. Abnormal expression of Claudin18.2 can occur in various primary malignant tumors, especially gastrointestinal tumors, and even in metastatic foci. It regulates its expression by activating the aPKC/MAPK/AP-1 pathway, and therefore, the Claudin18.2 protein is a pan-cancer target expressed in primary and metastatic lesions in human cancer types. Zolbetuximab (IMAB362), an antibody specific for Claudin18.2, has been successfully tested in a phase III clinical trial, and the results of the study showed that combining Zolbetuximab with chemotherapy notably extends patients' survival and is expected to be a potential first-line treatment for patients with Claudin18.2(+)/HER-2(-) gastric cancer. Here, we systematically describe the biological properties and oncogenic effects of Claudin18.2, centering on its clinical-pathological aspects and the progress of drug studies in gastric cancer, which can help to further explore its clinical value.

State of the art and upcoming trends in claudin-directed therapies in gastrointestinal malignancies.

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Claudins, components of tight cell junctions in epithelial and endothelial cells, have emerged as a therapeutic target in gastrointestinal (GI) malignancies, particularly claudin 18.2 (CLDN18.2).

Zolbetuximab, a chimeric anti-CLDN18.2 monoclonal antibody (mAb), is currently under FDA review and may emerge as the first claudin targeted therapy approved. Phase 3 trials show that zolbetuximab in combination with front-line fluoropyrimidine plus oxaliplatin improves survival in advanced CLDN18.2 positive ($\geq 75\%$ of tumor cells) gastric adenocarcinoma (GAC) patients. Many other therapies (mAbs; CART; bispecific; ADCs) are under investigation. CLDN18.2 will be an important target in GAC. Early understanding of how to target CLDN18.2 based on the level of expression (high, moderate, low) will be the key to success in this area. Studying these as separate entities should be considered. Resistance patterns, loss of CLDN18.2 expression, role in the refractory setting, and if any role in localized disease are questions that remain. Other targets for claudin that target claudin six and four are under investigation. Their role in GI malignancies will soon be further clarified.

New therapeutic target molecules for gastric and gastroesophageal junction cancer.

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Molecularly targeted therapy for receptor tyrosine kinases (RTKs) has faced limitations in gastric and gastroesophageal junction (G/GEJ) cancer except for HER2-targeted agents, possibly due to inappropriate assay selection that has hindered identification of sensitive patients, in addition to coexisting genetic abnormalities as well as intratumoral heterogeneity. Immunohistochemistry of RTKs has, thus, proved largely unsuccessful for patient selection, and detection of RTK gene amplification as a true oncogenic driver is problematic given the small numbers of affected individuals. FGFR2 amplification is associated with poor prognosis in G/GEJ cancer, and immunohistochemistry of the FGFR2b protein isoform has proved effective for the detection of such FGFR2-dependent tumors. Phase III and Ib/III trials of the FGFR2-targeted antibody bemarituzumab for G/GEJ cancer overexpressing FGFR2b are ongoing based on the promising result in a phase II trial, especially in cases with an FGFR2b positivity of $\geq 10\%$. Challenges to EGFR- and MET-targeted therapies are being tackled with antibody-drug conjugates (ADCs) and bispecific antibodies. CLDN18.2 is expressed in some G/GEJ tumors but lacks oncogenic driver potential, and the CLDN18.2-targeted antibody zolbetuximab prolonged the survival of CLDN18.2-positive G/GEJ cancer patients in phase III trials. Antibody-drug conjugates and ADCs that target CLDN18.2 are also being pursued for treatment of such patients. Similarly, targeting of nondriver molecules such as DKK1, TROP2, and CEACAM5 is under investigation in early-stage clinical trials. This shift in focus from target molecules with driver potential to markers for precise drug delivery should increase the number of possible targets in G/GEJ cancer.

Claudin 18.2 as a novel therapeutic target.

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Claudin 18.2, a tight-junction molecule predominantly found in the nonmalignant gastric epithelium, becomes accessible on the tumour cell surface during malignant transformation, thereby providing an appealing target for cancer therapy. Data from two phase III trials testing the anti-claudin 18.2 antibody zolbetuximab have established claudin 18.2-positive advanced-stage gastric cancers as an independent therapeutic subset that derives benefit from the addition of this agent to chemotherapy. This development has substantially increased the percentage of patients eligible for targeted therapy. Furthermore, newer treatments, such as high-affinity monoclonal antibodies, bispecific antibodies, chimeric antigen receptor T cells and antibody-drug conjugates capable of bystander killing effects, have

shown considerable promise in patients with claudin 18.2-expressing gastric cancers. This new development has resulted from drug developers moving beyond traditional targets, such as driver gene alterations or growth factors. In this Review, we highlight the biological rationale and explore the clinical activity of therapies that target claudin 18.2 in patients with advanced-stage gastric cancer and explore the potential for expansion of claudin 18.2-targeted therapies to patients with other claudin 18.2-positive solid tumours.

A novel highly sensitive inner filter effect-based fluorescence immunoassay with quantum dots for bioanalysis of zolbetuximab, a monoclonal antibody used for immunotherapy of gastric and gastroesophageal junction adenocarcinoma.

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Zolbetuximab (ZOL) is a groundbreaking monoclonal antibody targeting CLDN 18.2, a cancer cell surface protein. It is a first-in-class therapy for gastric and gastroesophageal junction adenocarcinoma. However, there is currently any immunoassay available for bioanalysis of ZOL, hindering its pharmacokinetic studies, therapeutic monitoring, and safety profile refinement. To address this gap, this study presents the development and validation of a novel highly sensitive inner filter effect-based fluorescence immunoassay (IFE-FIA) with quantum dots (QDs) as a probe. This assay enables the quantitative determination of ZOL in plasma samples. The assay involved non-competitive capturing of ZOL from the samples using a specific antigen (CLDN 18.2 protein) immobilized on assay plate microwells. A horseradish peroxidase (HRP)-labelled anti-human IgG was used to measure the immune complex. The assay's detection system relies on the formation of a light-absorbing colored product through an HRP-catalyzed oxidative reaction with the substrate 3,3',5,5'-tetramethylbenzidine. This light absorption efficiently quenched the fluorescence of QDs via the IFE. The measured fluorescence signals corresponded to the concentrations of ZOL in the samples. The conditions of the IFE-FIA and its detection system were refined, and the optimum procedures were established. Following the guidelines of immunoassay validation for bioanalysis, the assay was validated, and all the validation criteria were acceptable. The assay demonstrates high sensitivity, accurately quantifying ZOL at concentrations as low as 10 ng/mL in plasma samples, with acceptable precision. Importantly, it avoids interferences from endogenous substances and plasma matrix. The recoveries in spiked human plasma ranged from 96.8 % to 104.5 %, with relative standard deviations of 4.1 %-6.5 %. The proposed IFE-FIA represents a valuable tool for quantifying ZOL in clinical settings, enabling assessment of its pharmacokinetics, therapeutic drug monitoring, and safety profile refinement.

Efficacy and Safety of Zolbetuximab in Gastric Cancer.

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Gastric cancer remains a major global health concern with high incidence and mortality rates, particularly in East Asia. Patients often have poor outcomes due to limited treatment efficacy. Zolbetuximab, a monoclonal antibody targeting claudin 18.2 (CLDN18.2)-overexpressed in 50% to 80% of gastric cancers-demonstrates promise by initiating antibody-dependent cellular cytotoxicity and complement-dependent cytotoxicity in CLDN18.2-positive cells. In clinical trials, zolbetuximab with chemotherapy improved progression-free survival (PFS) and overall survival (OS). The FAST trial showed

a median OS increase from 8.4 months to 13.2 months (HR, 0.72; $P < .01$). The SPOTLIGHT trial found PFS extended to 11.0 months vs. 8.9 months (HR, 0.73; $P = .0024$) with OS reaching 18.2 months in the zolbetuximab arm. The GLOW trial also confirmed efficacy, with median OS improving from 12.16 months to 14.39 months (HR, 0.771; $P = .0118$). Zolbetuximab's targeted action, combined with manageable adverse effects, positions it as a promising therapy for advanced gastric cancer.

Claudin18.2 is a novel molecular biomarker for tumor-targeted immunotherapy.

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The claudin18.2 (CLDN18.2) protein, an isoform of claudin18, a member of the tight junction protein family, is a highly selective biomarker with limited expression in normal tissues and often abnormal expression during the occurrence and development of various primary malignant tumors, such as gastric cancer/gastroesophageal junction (GC/GEJ) cancer, breast cancer, colon cancer, liver cancer, head and neck cancer, bronchial cancer and non-small-cell lung cancer. CLDN18.2 participates in the proliferation, differentiation and migration of tumor cells. Recent studies have identified CLDN18.2 expression as a potential specific marker for the diagnosis and treatment of these tumors. With its specific expression pattern, CLDN18.2 has become a unique molecule for targeted therapy in different cancers, especially in GC; for example, agents such as zolbetuximab (claudiximab, IMAB362), a monoclonal antibody (mAb) against CLDN18.2, have been developed. In this review, we outline recent advances in the development of immunotherapy strategies targeting CLDN18.2, including monoclonal antibodies (mAbs), bispecific antibodies (BsAbs), chimeric antigen receptor T (CAR-T) cells redirected to target CLDN18.2, and antibody-drug conjugates (ADCs).
